Quasi-optimum distance flag codes

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Abstract

Given $\mathbb{F}_q$ a finite field, a *flag* of a vector space $\mathbb{F}_q^n$ is a sequence of nested subspaces. In network coding, a *flag code* is a collection of flags sharing the same sequence of dimensions. Here, we present the family of *quasi-optimum distance flag codes* as those attaining the second best possible distance value. We provide a characterization for them and exhibit upper bounds for their cardinality together with a proposal of systematic constructions for every choice of the type vector, based on *partial spreads* and *sunflowers*.

Key Words : Flag code, quasi-optimum distance, spread code, partial spread code, sunflower.

1 Introduction

Within the context of *network coding*, special attention has been given to *constant dimension codes*, which are those subspace codes in which all the codewords (subspaces) have the same dimension (see [EV11, TR18] and references therein). Important families of these codes, as the classes of *spread codes* and *partial spread codes*, introduced in [MGR08, GR14] respectively, are built from the classic notion of *spread* studied by Segre in [Seg64]. Another remarkable family is the one of *sunflowers*, families of subspaces of a given dimension, intersecting at a common subspace (see [ER15, GR16]).

Constant dimension codes have been generalized by the so called *multishot codes*, in which codewords are sequences of subspaces (see [NUF09]) and, in particular, by *flag codes*. The latter ones were introduced in [LNVC18] as sets of sequences of nested subspaces of $\mathbb{F}_q^n$, named *flags*, with prescribed *type vector* (sequence of dimensions). During the last years, several papers have been focused on the study of *optimum distance flag codes*, i.e., flag codes with the maximum possible distance (see [AGNPSE21b, AGNPSE20, AGNPSE21a, NPSE22]) as well as bounds for the cardinality of flag codes with a given value

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of the minimum distance ([AGNPSE23, Kur21]). In [AGNP24a], the authors presented a study of cyclic orbit flag codes attaining the "second best" value of the flag distance in the orbital scenario, that is, the so called *quasi-optimum distance cyclic orbit flag codes*. Recently, in [LYJ25] the authors study these codes in the general context for a specific choice of the type vector and present a construction based on MRD codes.

Flag codes have been also studied as a generalization of rank-metric codes. More precisely, in [ANZ24, FN23], the authors work with the *flag-rank distance*. In [ANZ24], the reader can find a Singleton-like bound for flag codes and constructions of *maximum flag-rank distance codes*. The notion of *quasi-optimum* flag codes, endowed with the flag-rank distance, already appears in that paper.

Here, we follow the ideas we have developed in [AGNP24b] and work with the usual flag distance to investigate general *quasi-optimum distance flag codes* focusing on a particular *distinguished projected flag code*, which contains, at most, four dimensions of the original type. Our characterization also allows us to upper-bound the cardinality of quasi-optimum distance flag codes, as well as to give a systematic construction for every type vector, mainly based on spreads, partial spreads and sunflowers.

Our results state a systematic way for the study and construction of flag codes attaining other controlled values of the distance.

2 Preliminares

Let q be a prime power and let $\mathbb{F}_q$ be the finite field with q elements. Consider $\mathbb{F}_q^n$ the n-dimensional vector space over $\mathbb{F}_q$ for $n \geqslant 2$ and $\mathcal{G}_q(k, n)$ the Grassmannian of dimension k endowed with *subspace distance* (see [KK08]):

$$d_S(\mathcal{U}, \mathcal{V}) = \dim(\mathcal{U} + \mathcal{V}) - \dim(\mathcal{U} \cap \mathcal{V}) = 2(k - \dim(\mathcal{U} \cap \mathcal{V})), \tag{1}$$

for every $\mathcal{U}, \mathcal{V} \in \mathcal{G}_q(k, n)$.

A *constant dimension code* (of dimension k) is a nonempty subset $\mathcal{C} \subset \mathcal{G}_q(k, n)$. Its minimum distance is given by the value

$$d_S(\mathcal{C}) = \min\{d_S(\mathcal{U}, \mathcal{V}) \mid \mathcal{U}, \mathcal{V} \in \mathcal{C},\ \mathcal{U} \neq \mathcal{V}\},$$

whenever $|\mathcal{C}| \geqslant 2$. In case $|\mathcal{C}| = 1$, we put $d_S(\mathcal{C}) = 0$. According to (1), the value $d_S(\mathcal{C})$ is an even integer such that $0 \leqslant d_S(\mathcal{C}) \leqslant \min\{2k, 2(n-k)\}$.

Proposition 2.1. *Let $\mathcal{C} \subseteq \mathcal{G}_q(k, n)$ with $k \leqslant \frac{n}{2}$. The code $\mathcal{C}$ has minimum distance $d_S(\mathcal{C}) = 2k$ if, and only if, every pair of different subspaces $\mathcal{U}, \mathcal{V} \in \mathcal{C}$ intersects trivially.*

Definition 2.2. A *partial k-spread* of $\mathbb{F}_q^n$ is a collection of k-dimensional vector spaces of $\mathbb{F}_q^n$ that intersect trivially. If the union of the elements in the partial spread is $\mathbb{F}_q^n$, then we speak about a *k-spread*.

These objects were originally studied in the context of finite geometry (see [Seg64]). In the framework of network coding, (partial) spread codes were introduced in [MGR08]. It is well-known that k-spreads exist if, and only if, k divides n and, in such a case, their cardinality equals $\frac{q^n-1}{q^k-1}$. In case k does not divide n, the size of partial k-spreads is upper bounded by

$$\left\lfloor \frac{q^n-1}{q^k-1} \right\rfloor \tag{2}$$

(see [GR14, Lemma 7] or [HKKW16, Th. 3.53]).

Proposition 2.3. *Consider a constant dimension code $\mathcal{C} \subseteq \mathcal{G}_q(k,n)$ with $k \geqslant \frac{n}{2}$. The code $\mathcal{C}$ has minimum distance $d_S(\mathcal{C}) = 2(n-k)$ (the maximum possible one) if, and only if, for every $\mathcal{U}, \mathcal{V} \in \mathcal{C}$, with $\mathcal{U} \neq \mathcal{V}$, we have $\mathcal{U} + \mathcal{V} = \mathbb{F}_q^n$.*

The knowledge of partial spread codes has also helped to provide constructions of codes with other prescribed values of the minimum distance. We will put special attention to the next family of codes.

Definition 2.4. A constant dimension code $\mathcal{C} \subseteq \mathcal{G}_q(k,n)$ is said to be a *sunflower of center $\mathcal{X}$* if, for every different subspaces $\mathcal{U}, \mathcal{V} \in \mathcal{C}$, we have $\mathcal{U} \cap \mathcal{V} = \mathcal{X}$.

In particular, the minimum distance of a sunflower of dimension k and center $\mathcal{X}$ is $2(k-c)$, with $c = \dim(\mathcal{X})$ (see [ER15, GR16]). In both papers [ER15, Th. 10 and Th. 11] and [GR16, Rem. 9], the authors proved the next result.

Theorem 2.5. *If $\mathcal{C}$ is a sunflower in $\mathcal{G}_q(k,n)$ of center $\mathcal{X}$ (of dimension c), then $\mathcal{C}$ is of the form*

$$\mathcal{C} = \mathcal{C}' \oplus \mathcal{X} = \{\mathcal{U}' \oplus \mathcal{X} \mid \mathcal{U}' \in \mathcal{C}'\},$$

where $\mathcal{C}'$ is a partial $(k-c)$-spread of $\mathbb{F}_q^{n-c}$.

This construction will be useful to provide constructions of flag codes with several prescribed values for the minimum distance using spreads, partial spreads and sunflowers.

2.1 Flag codes

Definition 2.6. Given integers $0 < t_1 < \cdots < t_r < n$, a *flag* of type $\mathbf{t} = (t_1, \ldots, t_r)$ on $\mathbb{F}_q^n$ is a sequence $\mathcal{F} = (\mathcal{F}_1, \ldots, \mathcal{F}_r)$ of $\mathbb{F}_q$-subspaces of $\mathbb{F}_q^n$ satisfying:

1. $\mathcal{F}_1 \subsetneq \cdots \subsetneq \mathcal{F}_r \subsetneq \mathbb{F}_q^n$ and
2. $\dim(\mathcal{F}_i) = t_i$, for all $1 \leqslant i \leqslant r$.

The set of all the flags of a given type vector $\mathbf{t} = (t_1, \ldots, t_r)$ is called the *flag variety* (of type $\mathbf{t}$) and we denote it by $\mathcal{F}_q(\mathbf{t}, n)$.

In this context, we consider error-correcting codes given by families of flags.

Definition 2.7. A *flag code* $\mathcal{C}$ of type $\mathbf{t} = (t_1, \ldots, t_r)$ on $\mathbb{F}_q^n$ is a nonempty subset of the flag variety $\mathcal{F}_q(\mathbf{t}, n)$.

Associated with any flag code, one can consider the following family of constant dimension codes.

Definition 2.8. Let $\mathcal{C}$ be a flag code of type $\mathbf{t} = (t_1, \ldots, t_r)$ on $\mathbb{F}_q^n$. For every $1 \leqslant i \leqslant r$, the *i-th projected (subspace) code* of $\mathcal{C}$ is the constant dimension code

$$\mathcal{C}_i = \{\mathcal{F}_i \mid \mathcal{F} \in \mathcal{C}\} \subseteq \mathcal{G}_q(t_i, n).$$

Similarly, we can consider the *projected codes of longer length* as follows. Given $2 \leqslant M \leqslant r$ indices $1 \leqslant i_1 \leqslant \ldots \leqslant i_M \leqslant r$, the *projected flag code* of length M and type $(t_{i_1}, \ldots, t_{i_M})$ of $\mathcal{C}$ is the set

$$\mathcal{C}_{(i_1,\ldots,i_M)} = \{(\mathcal{F}_{i_1}, \ldots, \mathcal{F}_{i_M}) \mid \mathcal{F} \in \mathcal{C}\} \subseteq \mathcal{F}_q((t_{i_1}, \ldots, t_{i_M}), n),$$

obtained by projecting onto the $(i_1, \ldots, i_M)$ coordinates of all the flags in $\mathcal{C}$.

In [AGNPSE20], the authors introduced the notion of disjointness for flag codes as follows.

Definition 2.9. A flag code $\mathcal{C}$ is said to be *disjoint* if $|\mathcal{C}| = |\mathcal{C}_1| = \cdots = |\mathcal{C}_r|$. In particular, in a disjoint flag code $\mathcal{C}$, different flags have different subspaces. On the other hand, if two different flags $\mathcal{F}, \mathcal{F}' \in \mathcal{C}$ have a common subspace $\mathcal{F}_i = \mathcal{F}'_i$ for some $1 \leqslant i \leqslant r$, we say that the flags $\mathcal{F}$ and $\mathcal{F}'$ *collapse* at dimension t_i.

The subspace distance in (1) can be extended to flags as follows. The *flag distance* between two flags $\mathcal{F}, \mathcal{F}'$ of type $\mathbf{t} = (t_1, \ldots, t_r)$ on $\mathbb{F}_q^n$ is computed as

$$d_f(\mathcal{F}, \mathcal{F}') = \sum_{i=1}^{r} d_S(\mathcal{F}_i, \mathcal{F}'_i) \tag{3}$$

and the *minimum distance* of a flag code $\mathcal{C}$ of type $\mathbf{t} = (t_1, \ldots, t_r)$ on $\mathbb{F}_q^n$ is

$$d_f(\mathcal{C}) = \min\{d_f(\mathcal{F}, \mathcal{F}') \mid \mathcal{F}, \mathcal{F} \in \mathcal{C},\ \mathcal{F} \neq \mathcal{F}'\},$$

whenever $|\mathcal{C}| \geqslant 2$. In case $|\mathcal{C}| = 1$, we simply put $d_f(\mathcal{C}) = 0$. The value $d_f(\mathcal{C})$ is an even integer satisfying $0 \leqslant d_f(\mathcal{C}) \leqslant D^{(\mathbf{t},n)}$, where

$$D^{(\mathbf{t},n)} = 2\left(\sum_{t_i \leqslant \frac{n}{2}} t_i + \sum_{t_i > \frac{n}{2}} (n - t_i)\right) = \sum_{i=1}^{r} \min\{2t_i, 2(n - t_i)\}. \tag{4}$$

Flag codes attaining this upper bound for the minimum distance are called *optimum distance flag codes* and they were characterized in [AGNPSE20] as follows.

Theorem 2.10. *([AGNPSE20, Th. 3.11]) Given a flag code $\mathcal{C} \subseteq \mathcal{F}_q((t_1, \ldots, t_r), n)$, they are equivalent:*

1. *$\mathcal{C}$ is an optimum distance flag code,*
2. *$\mathcal{C}$ is disjoint and $d_S(\mathcal{C}_i)$ is maximum for every $1 \leqslant i \leqslant r$.*

3 Quasi-optimum distance flag codes

Until now, the study of flag codes having a prescribed distance value has been mainly focused on those attaining the maximum possible distance. In this work, we address the next natural step and deal with general flag codes attaining the "second best" distance value, that is, those flag codes $\mathcal{C} \subseteq \mathcal{F}_q(\mathbf{t}, n)$ having *quasi-optimum distance*, that is, minimum distance $d_f(\mathcal{C}) = D^{(\mathbf{t},n)} - 2$.

Definition 3.1. Given the type vector $\mathbf{t} = (t_1, \dots, t_r)$, a flag code $\mathcal{C} \subseteq \mathcal{F}_q(\mathbf{t}, n)$ is said to be a *quasi-optimum distance flag code* (or QODFC, for short) if $d_f(\mathcal{C}) = D^{(\mathbf{t},n)} - 2$.

While optimum distance flag codes are always disjoint, QODFC do not need to satisfy this property. We characterize them by distinguishing two cases:

Disjoint codes

Take $\mathcal{C}$ a disjoint QODFC of type $\mathbf{t} = (t_1, \dots, t_r)$ on $\mathbb{F}_q^n$ and fix the following two dimensions (if they exist):

$$t_L = \max\{t_i \mid 2t_i \leqslant n\} \quad \text{and} \quad t_R = \min\{t_i \mid 2t_i \geqslant n\}. \tag{5}$$

Hence, the next result holds.

Theorem 3.2. *Let $\mathcal{C}$ be a disjoint flag code of type $\mathbf{t} = (t_1, \dots, t_r)$ on $\mathbb{F}_q^n$. Then $\mathcal{C}$ is a QODFC if, and only if, the following conditions hold:*

1. *$\mathcal{C}_i$ is a subspace code of maximum distance for every $i \in \{1, \dots, r\} \setminus \{L, R\}$,*
2. *$\mathcal{C}_{(L,R)}$ is a QODFC.*

Example 3.3. For $n = 5$ and $\mathbf{t} = (1, 2, 3)$, we have $t_L = 2$, $t_R = 3$ and $D^{(\mathbf{t},n)} = 2 + 4 + 4 = 10$. Consider flags $\mathcal{F}^1 = (\langle e_2 \rangle, \langle e_2, e_3 \rangle, \langle e_2, e_3, e_4 \rangle)$, $\mathcal{F}^2 = (\langle e_5 \rangle, \langle e_2, e_5 \rangle, \langle e_1, e_2, e_5 \rangle)$, $\mathcal{F}^3 = (\langle e_4 \rangle, \langle e_4, e_5 \rangle, \langle e_3, e_4, e_5 \rangle)$, where $\{e_1, \dots, e_5\}$ is the standard $\mathbb{F}_q$-basis of $\mathbb{F}_q^5$. The flag codes

$$\mathcal{C} = \{\mathcal{F}^1, \mathcal{F}^2\},\ \mathcal{C}' = \{\mathcal{F}^1, \mathcal{F}^3\},\ \mathcal{C}'' = \{\mathcal{F}^1, \mathcal{F}^2, \mathcal{F}^3\},$$

are QODFC, given that $d_f(\mathcal{C}) = d_f(\mathcal{C}') = d_f(\mathcal{C}'') = 8 = D^{(\mathbf{t},n)} - 2$. If we look at their projected codes, we observe that $d_S(\mathcal{C}_1) = d_S(\mathcal{C}_1') = d_S(\mathcal{C}_1'') = 2$ is maximum and

1. $d_S(\mathcal{C}_L) = 2$ and $d_S(\mathcal{C}_R) = 4$,
2. $d_S(\mathcal{C}_L') = 4$ and $d_S(\mathcal{C}_R') = 2$,
3. $d_S(\mathcal{C}_L'') = 2$ and $d_S(\mathcal{C}_R'') = 2$.

Moreover, one can easily check that

$$d_f(\mathcal{C}_{(2,3)}) = d_f(\mathcal{C}'_{(2,3)}) = d_f(\mathcal{C}''_{(2,3)}) = D^{((2,3),5)} - 2 = 6.$$

Actually, we need less projected subspace codes to determine if a flag code $\mathcal{C}$ is a QODFC.

Corollary 3.4. *Let $\mathcal{C}$ be a disjoint flag code of type $\mathbf{t} = (t_1, \dots, t_r)$ on $\mathbb{F}_q^n$. Then $\mathcal{C}$ is a QODFC if, and only if, the following conditions hold:*

1. *$\mathcal{C}_{L-1}, \mathcal{C}_{R+1}$ are subspace codes of maximum distance,*
2. *$\mathcal{C}_{(L,R)}$ is a QODFC.*

As a consequence, we can select a "central" projected code to check if a disjoint flag code $\mathcal{C}$ is a QODFC.

Definition 3.5. Let $\mathcal{C}$ be a flag code of type $\mathbf{t} = (t_1, \dots, t_r)$. The *distinguished projected code* of $\mathcal{C}$ (w.r.t. the value $D^{(\mathbf{t},n)} - 2$) is the flag code $\mathcal{C}_{(L-1,L,R,R+1)}$ of type $(t_{L-1}, t_L, t_R, t_{R+1})$, whenever these dimensions exist. Otherwise, we just ignore the dimensions not appearing in the original type vector.

Example 3.6. For $n = 10$, we consider different choices for the type vector $\mathbf{t}$. The next table shows the type vector of the corresponding distinguished projected flag code. We mark in boldface (resp. underlined) all the dimensions $k < \frac{n}{2}$ (resp. $k > \frac{n}{2}$) and we use the star symbol $\star$ when dimensions are not defined.

Type $\mathbf{t}$	t_L	t_R	$(t_{L-1}, t_L, t_R, t_{R+1})$	Distinguished type
$(\mathbf{1}, \mathbf{3}, \mathbf{4}, \underline{7}, \underline{8})$	4	7	$(\mathbf{3}, \mathbf{4}, \underline{7}, \underline{8})$	$(\mathbf{3}, \mathbf{4}, \underline{7}, \underline{8})$
$(\mathbf{1}, \mathbf{3}, \mathbf{4}, \underline{7})$	4	7	$(\mathbf{3}, \mathbf{4}, \underline{7}, \star)$	$(\mathbf{3}, \mathbf{4}, \underline{7})$
$(\mathbf{1}, \mathbf{3}, \mathbf{4})$	4	$\star$	$(\mathbf{3}, \mathbf{4}, \star, \star)$	$(\mathbf{3}, \mathbf{4})$
$(\mathbf{1}, \mathbf{3}, 5, \underline{7}, \underline{8})$	5	5	$(\mathbf{3}, 5, 5, \underline{7})$	$(\mathbf{3}, 5, \underline{7})$

Table 1: Distinguished type vectors for different $\mathbf{t} = (t_1, \dots, t_r)$

The previous results allow us to study QODFCs of any type vector looking at flag codes of length, at most, four.

Corollary 3.7. *Let $\mathcal{C}$ be a disjoint flag code of type $\mathbf{t} = (t_1, \dots, t_r)$ on $\mathbb{F}_q^n$. They are equivalent:*

1. *$\mathcal{C}$ is a QODFC and*
2. *the distinguished projected flag code $\mathcal{C}_{(L-1,L,R,R+1)}$ is a QODFC.*

Non-disjoint codes

Consider now non-disjoint QODFCs of type $\mathbf{t} = (t_1, \dots, t_r)$ on $\mathbb{F}_q^n$. In this case, we see that not every type vector is compatible with the distance $D^{(\mathbf{t},n)} - 2$.

Theorem 3.8. *Let $\mathcal{C}$ be a non-disjoint flag code in $\mathcal{F}_q(\mathbf{t}, n)$. Then $\mathcal{C}$ is a QODFC if, and only if, one of the following situations holds:*

1. *$t_1 = 1$, $\frac{n}{2} < t_2$ and $|\mathcal{C}_1| < |\mathcal{C}_2| = \cdots = |\mathcal{C}_r| = |\mathcal{C}|$. Moreover, $d_S(\mathcal{C}_i) = 2(n - t_i)$, for every $i \geqslant 3$, and $\mathcal{C}_{(1,2)}$ is a QODFC.*

2. $t_r = n-1$, $t_{r-1} < \frac{n}{2}$ *and* $|\mathcal{C}| = |\mathcal{C}_1| = \cdots = |\mathcal{C}_{r-1}| > |\mathcal{C}_r|$. *We also have* $d_S(\mathcal{C}_i) = 2t_i$ *for every* $i \leqslant r-2$, *and* $\mathcal{C}_{(r-1,r)}$ *is a QODFC.*

3. *The type vector is* $(1, n-1)$ *and* $|\mathcal{C}_i| < |\mathcal{C}|$, *for* $i \in \{1, 2\}$.

Remark 3.9. We can observe that, as it occurs in the disjoint case, for non-disjoint flag codes we have that the property of $\mathcal{C}$ being a QODFC is also satified by the projected flag code $\mathcal{C}_{(L,R)}$, where dimensions t_L and t_R are always well-defined and satisfy $t_L = 1$ and/or $t_R = n-1$. Each pair of different flags can present a collapse at their line or their hyperplane, but just at one of them.

In terms of a distinguished projected flag code of $\mathcal{C}$ we have:

Corollary 3.10. *Let* $\mathcal{C}$ *be a non-disjoint flag code in* $\mathcal{F}_q(\mathbf{t}, n)$. *Then* $\mathcal{C}$ *is a QODFC if, and only if, one of the following situations holds:*

1. $t_1 = 1$, $\frac{n}{2} < t_2$ *and* $|\mathcal{C}_1| < |\mathcal{C}_2| = |\mathcal{C}_3| = |\mathcal{C}|$. *Moreover,* $\mathcal{C}_3$ *is of maximum distance and* $\mathcal{C}_{(1,2)}$ *is a QODFC.*

2. $t_r = n-1$, $t_{r-1} < \frac{n}{2}$ *and* $|\mathcal{C}| = |\mathcal{C}_{r-2}| = |\mathcal{C}_{r-1}| > |\mathcal{C}_r|$. *We also have* $\mathcal{C}_{r-2}$ *is of maximum distance and* $\mathcal{C}_{(r-1,r)}$ *is a QODFC.*

3. *The type vector is* $(1, n-1)$ *and* $|\mathcal{C}_i| < |\mathcal{C}|$, *for* $i \in \{1, 2\}$.

Remark 3.11. Notice that there exist QODFCs $\mathcal{C}$ of type $(1, n-1)$ on $\mathbb{F}_q^n$ in which $|\mathcal{C}_1| = |\mathcal{C}|$ (resp. $|\mathcal{C}_2| = |\mathcal{C}|$). These cases are also taken into account in the previous result as a particular case of part (2) (resp. part (1)).

Corollary 3.12. *Let* $\mathcal{C}$ *be a non-disjoint flag code in* $\mathcal{F}_q(\mathbf{t}, n)$. *Then* $\mathcal{C}$ *is a QODFC if, and only if, one of the following situations holds:*

1. $t_1 = 1$, $\frac{n}{2} < t_2$ *and* $\mathcal{C}_{(1,2,3)}$ *is a QODFC.*

2. $t_r = n-1$, $t_{r-1} < \frac{n}{2}$ *and* $\mathcal{C}_{(r_2,r-1,r)}$ *is a QODFC.*

3. *The type vector is* $(1, n-1)$ *and* $|\mathcal{C}_i| < |\mathcal{C}|$, *for some* $i \in \{1, 2\}$.

Remark 3.13. The respective involved projected codes in the previous result form the corresponding distinguished projected flag code of $\mathcal{C}$: the first case corresponds to the type vector $(t_L = 1, t_R, t_{R+1})$; the second one to the case $(t_{L-1}, t_L, t_R = n-1)$; the last one holds for the type $(1, n-1) = (t_L, t_R)$.

4 Bounds for QODFC

Throughout this section, we will write $A_q(n, k, d)$ to denote the maximum possible size for a constant dimension code in $\mathcal{G}_q(k, n)$ with minimum distance d. For more information about these values, we refer the reader to the survey [HKKW16].

Theorem 4.1. *Let* $\mathcal{C}$ *be a disjoint QODFC of type* $\mathbf{t} = (t_1, \ldots, t_r)$. *Then it holds*

$$|\mathcal{C}| \leqslant \min\left\{ \left\lfloor \frac{q^n - 1}{q^{t_{L-1}} - 1} \right\rfloor, \left\lfloor \frac{q^n - 1}{q^{n-t_{R+1}} - 1} \right\rfloor \right\}.$$

Remark 4.2. If t_{L-1} or t_{R+1} does not appear in the type vector, then the previous result still holds true for the other one. If none of them is well-defined, we have type vectors of the form (t_L, t_R). In such a case, the next result follows.

Theorem 4.3. *Let $\mathcal{C}$ be a disjoint QODFC of type $\mathbf{t} = (t_L, t_R)$. Then*

$$|\mathcal{C}| \leqslant \min\{A_q(n, t_L, 2t_L - 2), A_q(n, t_R, 2(n - t_R) - 2)\}.$$

For the non-disjoint case, we consider three possibilities for the type vector.

Theorem 4.4. *Let $\mathcal{C}$ be a non-disjoint QODFC on $\mathbb{F}_q^n$.*

1. *If $\mathbf{t} = (1, t_2, \ldots, t_r)$ with $t_2 > \frac{n}{2}$, then*

$$|\mathcal{C}| \leqslant A_q(n, t_3, 2(n - t_3)) \leqslant \left\lfloor \frac{q^n - 1}{q^{n-t_3} - 1} \right\rfloor.$$

2. *If $\mathbf{t} = (t_1, \ldots, t_{r-1}, n - 1)$ with $t_{r-1} < \frac{n}{2}$, then*

$$|\mathcal{C}| \leqslant A_q(n, t_{r-2}, 2t_{r-2}) \leqslant \left\lfloor \frac{q^n - 1}{q^{t_{r-2}} - 1} \right\rfloor.$$

3. *Last, if $\mathbf{t} = (1, n - 1)$, then*

$$|\mathcal{C}| \leqslant |\mathcal{F}_q((1, n - 1), n)| = \frac{q^n - 1}{q - 1}\frac{q^{n-1} - 1}{q - 1}.$$

5 Constructions based on spreads and sunflowers

In this section we present a systematic construction of QODFC for any value of n and any type vector based on spreads, partial spreads and/or sunflowers. We divide our study into disjoint and non-disjoint flag codes.

5.1 The non-disjoint case

Type $\mathbf{t} = (t_1, \ldots, t_{r-1}, n - 1)$ with $t_{r-1} < \frac{n}{2}$

Consider a partial spread of dimension $t_{r-1} < \frac{n}{2}$ of $\mathbb{F}_q^n$ with size s

$$\mathcal{S} = \{\mathcal{S}_1, \ldots, \mathcal{S}_s\} \subseteq \mathcal{G}_q(t_{r-1}, n). \tag{6}$$

Notice that, for every pair of different subspaces $\mathcal{S}_i, \mathcal{S}_j \in \mathcal{S}$, there always exists a hyperplane $\mathcal{H}_{i,j}$ of $\mathbb{F}_q^n$ containing both $\mathcal{S}_i$ and $\mathcal{S}_j$.

Theorem 5.1. *Consider a type vector $\mathbf{t} = (t_1, \ldots, t_{r-1}, n - 1)$ with $t_{r-1} < \frac{n}{2}$ and let $\mathcal{S} \subseteq \mathcal{G}_q(t_{r-1}, n)$ be the partial spread in (6). Every flag code $\mathcal{C} = \{\mathcal{F}^1, \ldots, \mathcal{F}^s\}$ satisfying the conditions:*

1. *$\mathcal{F}_{r-1}^i = \mathcal{S}_i$, for every $1 \leqslant i \leqslant s$ and*
2. *$\mathcal{F}_r^1 = \mathcal{F}_r^2 = \mathcal{H}_{1,2} \in \mathcal{G}_q(n - 1, n)$ such that $\mathcal{S}_1 + \mathcal{S}_2 \subseteq \mathcal{H}_{1,2}$.*

is a QODFC of type $\mathbf{t}$ and size $|\mathcal{C}| = s$ with projected code $\mathcal{C}_{r-1} = \mathcal{S}$.

Type $\mathbf{t} = (1, t_2, \ldots, t_r)$ with $t_2 > \frac{n}{2}$

This situation can be reduced to the previous case by duality arguments.

Type $\mathbf{t} = (1, n-1)$

In this case, the whole flag variety itself is the largest possible QODFC.

Theorem 5.2. *The code $\mathcal{C} = \mathcal{F}_q((1, n-1), n)$ is a QODFC with maximum size.*

5.2 The disjoint case

We can assume that our type vector $\mathbf{t}$ contains some dimension $t_i \leqslant \frac{n}{2}$ and distinguish two cases: either $n = 2k+1$ or $2k+2$ for a positive integer $k \geqslant 1$. In any case, we start our construction by fixing an arbitrary subspace $\mathcal{X} \in \mathcal{G}_q(2k, n)$ and a k-spread $\mathcal{S} = \{\mathcal{S}_1, \ldots, \mathcal{S}_{q^k+1}\}$ of $\mathcal{X}$. For every $1 \leqslant i \leqslant q^k+1$, we choose a full-rank matrix $S_i \in \mathbb{F}_q^{k \times n}$ whose rows span the corresponding $\mathcal{S}_i$, that is, $\mathcal{S}_i = \mathrm{rowsp}(S_i)$. Moreover, for every $1 \leqslant j \leqslant k$, we denote by $S_i^{(j)}$ the matrix given by the first j rows of S_i and consider the subspace $\mathcal{S}_i^{(j)} = \mathrm{rowsp}(S_i^{(j)})$. With this notation, and for every $1 \leqslant j \leqslant k$, we have the next partial spread:

$$\mathcal{S}^{(j)} = \{\mathcal{S}_1^{(j)}, \ldots, \mathcal{S}_{q^k+1}^{(j)}\}. \tag{7}$$

Take now a vector $u \in \mathbb{F}_q^n \setminus \mathcal{X}$ and, for every $1 \leqslant j < \frac{n}{2}$, we form the set of subspaces:

$$\mathcal{S}^{(j)} \oplus \langle u \rangle = \{\mathcal{S}_i^{(j)} \oplus \langle u \rangle \mid 1 \leqslant i \leqslant q^k+1\}. \tag{8}$$

By means of Theorem 2.5, the next result follows.

Theorem 5.3. *For every $1 \leqslant j < \left\lfloor \frac{n}{2} \right\rfloor$, the code $\mathcal{S}^{(j)} \oplus \langle u \rangle$ given in (8) is a sunflower of dimension $j+1 \leqslant \left\lfloor \frac{n}{2} \right\rfloor$ of $\mathbb{F}_q^n$ and center $\langle u \rangle$. In particular, $|\mathcal{S}^{(j)} \oplus \langle u \rangle| = q^k+1$ and its minimum distance $d_S(\mathcal{S}^{(j)} \oplus \langle u \rangle) = 2j$ is the "second best" possible one for dimension $j+1$.*

For higher dimensions, we present the following families of constant dimension codes of $\mathbb{F}_q^n$ with maximum distance. If $n = 2k+1$, for every $1 \leqslant j \leqslant k$, we consider subspaces

$$\left\{ \begin{array}{lcl} \mathcal{U}_1^{(j)} & = & \mathcal{S}_1 \oplus \mathcal{S}_2^{(j-1)} \oplus \langle u \rangle, \\ \mathcal{U}_2^{(j)} & = & \mathcal{S}_2 \oplus \mathcal{S}_3^{(j-1)} \oplus \langle u \rangle, \\ \vdots & \vdots & \vdots \\ \mathcal{U}_{q^k+1}^{(j)} & = & \mathcal{S}_{q^k+1} \oplus \mathcal{S}_1^{(j-1)} \oplus \langle u \rangle, \end{array} \right. \tag{9}$$

of dimension $k+j \in \{k+1, \ldots, 2k = n-1\}$ and we form the subspace code

$$\mathbf{C}^{(j)} = \{\mathcal{U}_1^{(j)}, \ldots, \mathcal{U}_{q^k+1}^{(j)}\} \subseteq \mathcal{G}_q(k+j, 2k+1) \tag{10}$$

of dimension $k+j \geqslant k+1 > \frac{n}{2}$.

If $n = 2k + 2$, we consider an additional vector $v \in \mathbb{F}_q^n \setminus (\mathcal{X} \oplus \langle u \rangle)$ and, for every $1 \leqslant j \leqslant k$, we take subspaces

$$\begin{cases} \mathcal{V}_1^{(j)} & = & \mathcal{S}_1 \oplus \mathcal{S}_2^{(j-1)} \oplus \langle u, v \rangle, \\ \mathcal{V}_2^{(j)} & = & \mathcal{S}_2 \oplus \mathcal{S}_3^{(j-1)} \oplus \langle u, v \rangle, \\ \vdots & \vdots & \vdots \\ \mathcal{V}_{q^k+1}^{(j)} & = & \mathcal{S}_{q^k+1} \oplus \mathcal{S}_1^{(j-1)} \oplus \langle u, v \rangle, \end{cases} \tag{11}$$

all of dimension $k + j + 1 \in \{k + 2, \ldots, 2k + 1 = n - 1\}$ of $\mathbb{F}_q^n$ and construct the code

$$\mathbf{D}^{(j)} = \{\mathcal{V}_1^{(j)}, \ldots, \mathcal{V}_{q^k+1}^{(j)}\} \subseteq \mathcal{G}_q(k + j + 1, 2k + 2). \tag{12}$$

with dimension $k + j + 1 \geqslant k + 2 > k + 1 = \frac{n}{2}$. With this notation, we have:

Theorem 5.4. *For every $1 \leqslant j \leqslant k$, both codes $\mathbf{C}^{(j)} \subseteq \mathcal{G}_q(k + j, 2k + 1)$ and $\mathbf{D}^{(j)} \subseteq \mathcal{G}_q(k+j+1, 2k+2)$ defined in (10) and (12), respectively, have maximum distance.*

We can now give our construction of QODFC: consider the next family of flags of type $\mathbf{t} = (t_1, \ldots, t_r)$, having at least a dimension $t_L \leqslant \frac{n}{2}$.

For $n = 2k + 1$, and for every $1 \leqslant i \leqslant q^k + 1$, we consider the flag $\mathcal{F}^i = (\mathcal{F}_1^i, \ldots, \mathcal{F}_r^i)$ with subspaces

$$\mathcal{F}_j^i = \begin{cases} \mathcal{S}_i^{(t_j)} & \text{if} \quad 1 \leqslant j \leqslant L - 1, \\ \mathcal{S}_i^{(t_j-1)} \oplus \langle u \rangle, & \text{if} \quad j = L, \\ \mathcal{U}_i^{(t_j-k)} = \mathcal{S}_i \oplus \mathcal{S}_{i+1}^{(t_j-k-1)} \oplus \langle u \rangle & \text{if} \quad L < j \leqslant r, \end{cases} \tag{13}$$

Similarly, if $n = 2k + 2$, we define flags $\mathcal{F}^i$ with subspaces:

$$\mathcal{F}_j^i = \begin{cases} \mathcal{S}_i^{(t_j)} & \text{if} \quad 1 \leqslant j \leqslant L - 1, \\ \mathcal{S}_i^{(t_j-1)} \oplus \langle u \rangle, & \text{if} \quad j = L, \\ \mathcal{V}_i^{(t_j-k-1)} = \mathcal{S}_i \oplus \mathcal{S}_{i+1}^{(t_j-k-2)} \oplus \langle u, v \rangle & \text{if} \quad L < j \leqslant r, \end{cases} \tag{14}$$

In any case, we consider the flag code

$$\mathcal{C} = \{\mathcal{F}^1, \ldots, \mathcal{F}^{q^k+1}\} \subseteq \mathcal{F}_q((t_1, \ldots, t_r), n) \tag{15}$$

with flags defined in (13) or (14) depending on the parity of $n \in \{2k+1, 2k+2\}$.

Theorem 5.5. *The flag code $\mathcal{C}$ given in (15) is a disjoint QODFC of type $\mathbf{t} = (t_1, \ldots, t_r)$ on $\mathbb{F}_q^n$ with size $q^k + 1$.*

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